



Geographic Information Systems in R: Fundamentals and Applications for Ecologists

CONS 6970DL-4 & SMSC 524-DL2

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Course Overview

This asynchronous online course aims to provide graduate students with an introduction to the increasingly powerful and flexible range of tools for working with both vector and raster spatial data in the R environment. The course will review core GIS concepts and the basics of R programming and build skills in raster and vector import and manipulation. The course closely integrates tidyverse functions with GIS tools to develop efficient and intuitive coding. Special emphasis will be placed on visualizing spatial data by developing static and interactive maps. The first few weeks will be spent honing R coding skills, ensuring that you have the necessary understanding of modern data science workflows in R.

Credits: 3

Prerequisites: Prior experience with the R coding environment, for example reading in csv files.

Course Learning Outcomes

By the end of this course, students will be able to:

- Use tidyverse functions for data manipulation and visualization
- Read, project, and write shapefiles and raster files in R
- Conduct spatial joins, geometric operations, and data extraction with geospatial files
- Identify and interpret coordinate reference systems and projections and transform spatial data between them
- Conduct geospatial analyses (e.g., point pattern analysis)
- Create informative and visually appealing static and interactive maps
- *Optional:* Create interactive web applications for exploring geospatial data

Format and Pacing

As an asynchronous course, there is no set mandatory meeting time. However, students **are** required to submit weekly problem sets by a specified due date. Aside from this requirement, students may view the instructional videos and work through associated exercises (see below) at their own pace.

Class Meeting Days, Times, and Modalities

Weekly virtual meetings

There will be two *optional* live weekly virtual sessions on Zoom. For those who learn better by interacting with others, this “virtual classroom” provides an opportunity to do so. *Note: Times are reported in Eastern Time (ET) and Coordinated Universal Time (UTC).*

- **Review session** (Mondays, 10:30–11:45 ET [14:30–15:45 UTC]; [join the review session at this link](#)): Review of the previous week’s assignment, led by the instructor. Answer keys to the problem sets will be posted each Monday by 00:01 ET (04:01 UTC), giving participants time to review the key before the Monday review session.
- **Question & Answer session** (Fridays, 10:45–11:30 ET [14:45–15:30 UTC]; [join the Question & Answer session at this link](#)): an open question-and-answer period. During this session, there will be time to ask questions about lecture content and the current week’s assignment. Questions will be addressed on a first-come, first-served basis. A discussion-board thread will be used to submit questions before the meeting, though participants may also ask questions orally during the session.

Note: Both meetings will be recorded and posted on Canvas for those who are unable to attend.

Virtual office hours

Office hours (Zoom) provide an opportunity for students to interact one-on-one with instructors. Office hours are by appointment only ([sign up for office hours at this link](#)) and each office-hour time slot will be 15 minutes. Students must sign up for time slots a minimum of 24 hours prior to the office-hours session and no more than one week in advance.

- Thursdays (08:00–09:30 ET [12:00–13:30 UTC]; [join Thursday office hours at this link](#))
- Fridays (13:30–14:45 ET [17:30–18:45 UTC]; [join Friday office hours at this link](#))

Learning Content and Course Tools

Instructional Videos and Exercises

Learning content will be provided via weekly “learning modules” composed of HTML tutorials, instructional videos, commented R scripts, and self-assessment exercises. A new module will be posted every Monday morning (ET) on Canvas. The total length of instructional videos will vary by week but is generally in the range of 1–1.75 hours. Students should expect to devote approximately 4–8 hours per week to this content (Note: this estimate does not include time spent on assignments). The time required may be more or less, depending on your pace when working through exercises and your previous experience with R.

Pre-course Exercises

Preliminary exercises in R will be provided at least one week prior to the course and are required for all participants. The lessons are designed to help R beginners develop basic familiarity with R programming concepts and will serve an important refresher for intermediate R users.

Readings

Optional readings will be listed in the introduction section of each module and are intended for students who would like to explore each topic in greater depth.

Communication

Announcements

Announcements and reminders posted on Canvas will be the primary form of communication for material relevant to the whole class. ***For personal communication, email is the best means to reach us.*** Emails received during the work week will be returned within 24–48 hours. Emails received over the weekend may not be returned until the following Monday.

Online Discussion Board

A discussion board will be hosted on Canvas for students to discuss the week's assignment, seek advice on approaches to a given question, and provide mutual assistance. We recommend engaging with the discussion board often — it is a great place to find help with a problem or to help a classmate! We will monitor the discussion board frequently, but we predominantly consider this a peer-to-peer learning opportunity and will only comment when necessary.

Student and Instructor Expectations

Time Expectations

Depending on previous experience with R, students should expect to spend 4–8 hours per week on instructional content, interacting on discussion boards, and virtual meetings. Time spent completing the readings and problem sets is not included in this estimate but is expected to require an additional 8–10 hours per week. Note that for a 3-credit, half-semester course, George Mason University's suggested time expectation for students is 18 hours per week.

Computing Requirements

Students will be required to conduct all analyses in R using RStudio; both are free and open source. Students are expected to install these programs on their personal computers.

Evaluation

Problem Sets

Weekly Problem Sets: During weeks 1–7, we will post weekly problem sets to Canvas on Monday mornings. Problem sets are due the following Sunday by 23:59 ET (03:59 UTC the following day) and will be submitted through Canvas. Assignments will be posted as HTML and R script files. Because problem set solutions will be posted immediately after the submission window is closed, late assignments will not be accepted. *Please email the instructors if emergency circumstances require an exception.*

Final Problem Set: In addition to the weekly assignments, students will complete a final problem set. This open-book exercise will evaluate your ability to integrate content across learning modules. The problem set will focus on wrangling a real-world dataset and generating

useful data products using those data. Students will be provided with a dataset and will receive an HTML file and an R script with prompts for necessary wrangling steps, analysis, and output. The final week of the course will be devoted to this work.

Assignment Expectations and Submission: All problem sets consist of a series of questions in an HTML document and an R script. Each assignment will be submitted as a well-organized and neatly formatted R script with any relevant code. You will be provided with a list of functions that may be used to complete the assignment — only the provided functions may be used. The points allotted to each question will be provided in the HTML document. Problem sets will be returned no later than three days after they are due.

Individual Work: Collaboration is encouraged and will be highly beneficial in completing the problem sets. However, each student is expected to produce their own code and assignment submission. When posting to the discussion board, keep this in mind: providing support or answering a question is different from providing the answer to a problem.

Use of AI (Artificial Intelligence) Tools: You are permitted to use generative AI to explore ideas, explain errors, or review your submission, but please do not submit AI-generated code. AI works great as a thought partner but does not replace informed decision-making! In our experience, AI-generated R guidance often relies on outdated workflows — for example, deprecated functions or code that does not account for newer function arguments.

Participation

For-credit students will receive a participation score based on their contributions to the course discussion board across weeks. On a given week, a student's participation grade will be "1" if they posted to the discussion board or responded to a post, or "0" if they did not.

Certificate Requirements

Those interested in receiving a formal certificate must earn at least 70% on at least three weekly assignments and at least 70% on the final problem set. Participants not requiring a certificate are not required to submit anything, though they may certainly do so. Full answer keys will be provided, and multiple sessions are available for asking and discussing questions.

Grading

Students taking this course for university credit will receive a letter grade, assigned as:

A	≥ 93%	C+	≥ 77% and < 80%
A-	≥ 90% and < 93%	C	≥ 73% and < 77%
B+	≥ 87% and < 90%	C-	≥ 70% and < 73%
B	≥ 83% and < 87%	D	≥ 60% and < 70%
B-	≥ 80% and < 83%	F	< 60%

Final course grades will be based on:

Weekly Problem Sets (6 highest-scoring submissions):	60%
Final Problem Set:	30%
Participation (proportion of weeks with a discussion-board post or response):	10%

University Policies

Academic Standards

Academic Standards exist to promote authentic scholarship, support the institution's goal of maintaining high standards of academic excellence, and encourage continued ethical behavior of faculty and students to cultivate an educational community which values integrity and produces graduates who carry this commitment forward into professional practice.

As members of the George Mason University community, we are committed to fostering an environment of trust, respect, and scholarly excellence. Our academic standards are the foundation of this commitment, guiding our behavior and interactions within this academic community. The practices for implementing these standards adapt to modern practices, disciplinary contexts, and technological advancements. Our standards are embodied in our courses, policies, and scholarship, and are upheld in the following principles:

- **Honesty:** Providing accurate information in all academic endeavors, including communications, assignments, and examinations.
- **Acknowledgement:** Giving proper credit for all contributions to one's work. This involves the use of accurate citations and references for any ideas, words, or materials created by others in the style appropriate to the discipline. It also includes acknowledging shared authorship in group projects, co-authored pieces, and project reports.
- **Uniqueness of Work:** Ensuring that all submitted work is the result of one's own effort and is original, including free from self-plagiarism. This principle extends to written assignments, code, presentations, exams, and all other forms of academic work.

Violations of these standards — including but not limited to plagiarism, fabrication, and cheating — are taken seriously and will be addressed in accordance with university policies. The process for reporting, investigating, and adjudicating violations is [outlined in the university's procedures](#). Consequences of violations may include academic sanctions, disciplinary actions, and other measures necessary to uphold the integrity of our academic community.

The principles outlined in these academic standards reflect our collective commitment to upholding the highest standards of honesty, acknowledgement, and uniqueness of work. By adhering to these principles, we ensure the continued excellence and integrity of George Mason University's academic community.

Student responsibility: Students are responsible for understanding how these general expectations regarding academic standards apply to each course, assignment, or exam they participate in; students should ask their instructor for clarification on any aspect that is not clear to them.

Accommodations for Students with Disabilities

Disability Services at George Mason University is committed to upholding the letter and spirit of the laws that ensure equal treatment of people with disabilities. Under the administration of University Life, Disability Services implements and coordinates reasonable accommodations and disability-related services that afford equal access to university programs and activities. Students can begin the registration process with Disability Services at any time during their enrollment at George Mason University. If you are seeking accommodations, please visit <https://ds.gmu.edu/> for detailed information about the Disability Services registration

process. Disability Services is located in Student Union Building I (SUB I), Suite 2500. Email: ods@gmu.edu. Phone: (703) 993-2474.

Student responsibility: Students are responsible for registering with Disability Services and communicating about their approved accommodations with their instructor *in advance* of any relevant class meeting, assignment, or exam.

FERPA and Use of GMU Email Addresses for Course Communication

The [Family Educational Rights and Privacy Act \(FERPA\)](#) governs the disclosure of [education records for eligible students](#) and is an essential aspect of any course. **Students must use their GMU email account** to receive important University information, including communications related to this class. Instructors will not respond to messages sent from or send messages regarding course content to a non-GMU email address.

Student responsibility: Students are responsible for checking their GMU email regularly for course-related information, and/or ensuring that GMU email messages are forwarded to an account they do check.

Title IX and Non-Discrimination: Access and Compliance with George Mason's Nondiscrimination Policies

George Mason University is committed to providing a safe and non-discriminatory learning, living, and working environment across all educational programs, services, and activities for all members of the University community. University educational programs, services, and activities will be free from discrimination on the basis of race, color, religion, ethnic national origin (including shared ancestry and/or ethnic characteristics), sex, disability, military status (including veteran status), sexual orientation, gender identity, gender expression, age, marital status, pregnancy status, genetic information, or any other characteristic protected by law.

Required Reporting

All non-confidential employees, including your faculty members, have a legal requirement to report all relevant details, obtained directly or indirectly, about any incident of discrimination or harassment, including Title IX Prohibited Conduct, to the Office of Access, Compliance, and Community (OACC), which includes the Title IX Coordinator.

Reportable Conduct by an employee, a student, or a third party includes:

- Discrimination and harassment under Title VI or Title VII, which can include unequal and unfavorable treatment in the educational/work environment or conduct (oral, written, graphic) that is physically threatening, harmful, or humiliating.
- Prohibited Conduct under Title IX, which can include sexual harassment, sexual assault, gender-based stalking, or dating/domestic violence.

Upon notification, OACC professionals will assess the report and determine if outreach is required. If outreach is required, the potentially harmed individual (the "Complainant") will receive a communication, likely in the form of an email, with an offer to meet with a representative of the appropriate team.

Applicable Policies and Resources

- [University Policy 1201: Non-Discrimination](#)

- Contact masoneo@gmu.edu with related questions.
- [University Policy 1202: Sexual and Gender-Based Harassment and Other Interpersonal Violence](#)
 - Contact titleix@gmu.edu with related questions.
- [University Policy 1204: Prohibited Relationships with Students](#)

General questions may be directed to OACC at oacc@gmu.edu, or by phone at 703-993-8730, or in person on the Fairfax campus in Aquia 373 or at Mason Square in Van Metre Hall Suite 222.

Student opportunity: If you prefer to speak to someone *confidentially*, please contact one of Mason's confidential employees in [Student Support and Advocacy \(SSAC\)](#), [Counseling and Psychological Services \(CAPS\)](#), [Student Health Services \(SHS\)](#), and/or the [Office of the University Ombudsperson](#).